

# RADIATION AND DARCY EFFECTS ON UNSTEADY MHD HEAT AND MASS TRANSFER FLOW OF A CHEMICALLY REACTING FLUID PAST AN IMPULSIVELY STARTED VERTICAL PLATE WITH HEAT GENERATION

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## ABSTRACT

This paper is focused on the effects of radiation and mass transfer on a laminar free convective flow of a viscous, incompressible, electrically conducting and chemically reacting fluid past an impulsively started moving vertical plate adjacent to Darcian porous regime in the presence of heat generation. The governing boundary-layer equations are formulated in an  $(x, y, t)$  coordinate system with appropriate boundary conditions. The Rosseland diffusion approximation is used to analyze the radiative heat flux in the energy equation, which is appropriate for non-scattering media. The dimensionless governing equations are solved using an implicit finite-difference method of Crank-Nicolson type. A parametric study is performed to illustrate the influence of thermophysical parameters on the velocity, temperature and concentration profiles. Also, the local and average skin-friction, Nusselt number and Sherwood number are presented graphically. It is found that thermal radiation reduces both velocity and temperature in the boundary layer. Increasing the Darcy number is seen to accelerate the flow. An increase in the chemical reaction parameter leads a fall in the concentration. This model finds applications in solar energy collection systems, porous combustors and transport in fires in porous media (forest fires) and also in the design of high temperature chemical processing systems.

**Keywords:** Radiation, MHD, Darcian regime, Heat generation, Chemical reaction

## 1 INTRODUCTION

The role of thermal radiation is a major importance in some industrial applications such as glass production and furnace design and in space technology applications, such as cosmoical flight aerodynamics rocket, propulsion systems, plasma physics and space craft reentry. In the processes involving high temperatures, the radiation heat transfer in combination with conduction, convection and also mass transfer plays very important role in the design of pertinent equipments in the areas such as nuclear power plants, gas turbines and the various propulsion devices for air craft, missiles,